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# Introduction to Computer Security

# What is Computer Security? (1 of 3)



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It is the “Art & Science” of protecting or securing  
computer systems

# What is Computer Security? (2 of 3)



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It is the “Art & Science” of protecting/securing computer systems

- What does protecting or securing mean?
  - Examples:
    - protecting your computer against viruses or other malware
    - protecting your information (e.g., tax returns) from being taken or accessed by others (*unauthorized*)



# What is Computer Security? (3 of 3)



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It is the “Art & Science” of protecting/securing computer systems?

- Why is it considered both “Art and Science”
  - while many things in computer security are science or engineering, some aspects remain an ‘Art’
  - Examples: security mindset, designing ciphers, etc

# Why do we need computer security? (1 of 2)



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Because we **need** to protect or secure computers  
against .... adversaries



# Why do we need computer security? (2 of 2)



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Because we **need** to protect or secure computers against .... adversaries

- Who are these *adversaries*?
  - Prying family/friends (?) 😊
  - Mischief makers (script kiddies)
  - Hackers (fame/money/...)
  - Hacktivists (hackers but motivated by a cause)
  - Organized crime
  - Nation states
  - ...

# Why is computer security challenging?



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- Both systems to be protected and security mechanisms can be quite complex and subtle
- Security mechanisms themselves might become targets or introduce unintended weaknesses
- A single weakness can bring down the system – defenders have to work harder
- Systems, environments, and adversaries are constantly evolving/changing
- Security often tends to be an afterthought rather than designed in
- Users play a crucial role and usable security is hard
- ....

# Summary



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- Computer security is the art and science of protecting computer systems against potential adversaries
- Computer security is challenging because of ever changing systems, environments and adversaries, and complexity of systems among other things.

# Key Security Notions/Attributes

# Confidentiality



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- Preventing unauthorized access or disclosure of data/information
  - Keeping data (and metadata) confidential to authorized parties
- Example:
  - if corporate spies or outsiders gain access to coca-cola's formula that would be a serious loss of confidentiality

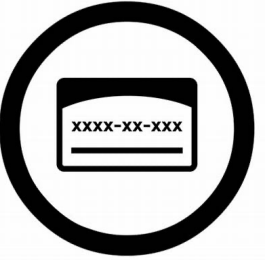


# Privacy



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- Preventing unauthorized access or disclosure of information/data **about people**
- An individual's right to decide who gets access to information/data about them
- Examples:
  - if attackers breach Target to steal credit card numbers from Target or breach IRS to steal SSN that would be a violation of privacy
  - if a hospital shares your medical information with others without your permission that would be a breach of privacy





- Preventing against unauthorized modifications
  - Data Integrity (integrity)
  - Origin Integrity (authentication)
  - System Integrity
- Examples:
  - overwriting your bank balance file/database
  - forged from address in e-mail, junk call, IP packet
  - malware on computer (unauthorized change to system)

# Availability



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- Ensuring timely availability of (data, system service etc.)
- Example:
  - Denial-of-service attack taking down a network/server

# Authenticity



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- Property of being genuine; can be verified and trusted
  - How is this related to integrity?
- Example:
  - A pirated copy of software

# Accountability



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- Requirement for entity actions to be traced uniquely to that entity
- Non-repudiation
  - one cannot repudiate one's actions
- Example:
  - paying with a credit card and later claiming never having done the transaction

# Summary



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- Key Security Attributes:
  - Confidentiality
  - Integrity
  - Availability
  - Privacy
  - Authenticity
  - Accountability
- Confidentiality, Integrity and Availability are remembered as 'CIA'
- These notions are not all independent
  - Confidentiality or Secrecy <-> Privacy
  - Integrity <-> Authenticity



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# **Some Computer Security Terminology**

# Asset



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- Something of value
- Examples:
  - data, system, service, networks, hardware, compute cycles etc.

# Threat



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- Set of circumstances that has the potential to breach security and cause harm
- Example:
  - exposure of customer data
  - corruption of database

# Some common threats



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- Snooping or interception
  - Unauthorized interception of information
- Falsification
  - Unauthorized change of information
- Masquerading or spoofing
  - An impersonation of one entity by another
- Repudiation
  - A false denial of receiving some information or taking an action.

# Are all threats/attacks equally important?



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- Low Impact
  - Limited adverse impact – e.g., loss of time/productivity
- Moderate Impact
  - Serious adverse impact – e.g., some loss of revenue, loss of capability
- High Impact
  - Severe or catastrophic – e.g., potentially bankrupting, loss of life or physical harm

# Adversary



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- Threat agent or an entity that materializes the threat
- Examples:
  - a hackers group that is targeting your customer data

# Vulnerability



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- A weakness in the system that could potentially be exploited to violate security property of interest
- Examples:
  - A misconfiguration or weak security configuration in a computer
  - A bug in the software that could be exploited; like Heartbleed in OpenSSL
  - Weak passwords

# Attack



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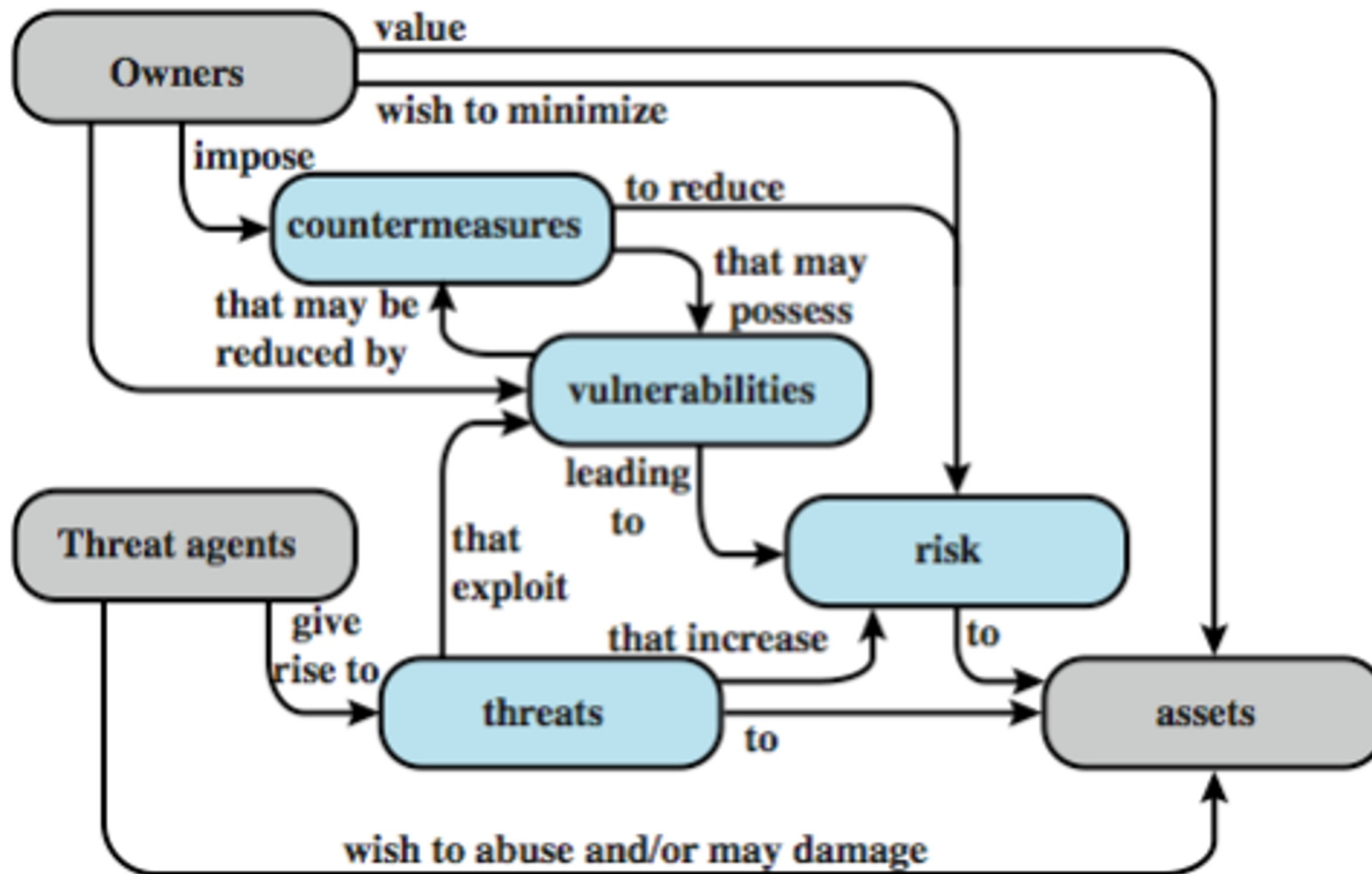
- When an entity exploits a vulnerability in a system to violate a security property/attribute
- Examples:
  - an attacker exploits heartbleed bug to steal your cryptographic keys
  - an attacker guesses your weak password to take over your account

# Control or Countermeasure



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- A means or method to
  - prevent a vulnerability from being exploited;
  - or minimize harm from the vulnerability/attack;
  - or detect attack so response and recovery actions may be initiated
  - or respond to or recover from an attack
- Examples:
  - Storing sensitive data encrypted to mitigate the impact of data being stolen
  - An independent data back-up system to prevent loss of data from failures and ransomware attacks



# Attack Surface



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- Is it the sum of vulnerabilities?
- Is it the sum of all exploitable vulnerabilities?
- Reachable exploitable vulnerabilities
  - Network
  - Software
  - Human

# Attack Surface Categories



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## Network

- Vulnerabilities accessible over the network
- Include network protocol vulnerabilities

## Software

- Vulnerabilities in application, utility, or operating system code
- Web server software in particular

## Human

- Social engineering
- Phishing

# Summary



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- We learned the following security terms and relations among them
  - Asset
  - Threat
  - Adversary
  - Vulnerability
  - Controls/Countermeasures
  - Attack Surface



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# Cyber Security Principles

# Security Principles



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Economy of Mechanism (Keep it simple)

Fail-safe (fail-close)

Complete Mediation

Open Design

Separation-of-privilege

Least-privilege

Least common mechanism

Psychological Adoptability (Usability)

# Security Principles



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Work Factor

Failure Recording or Detection

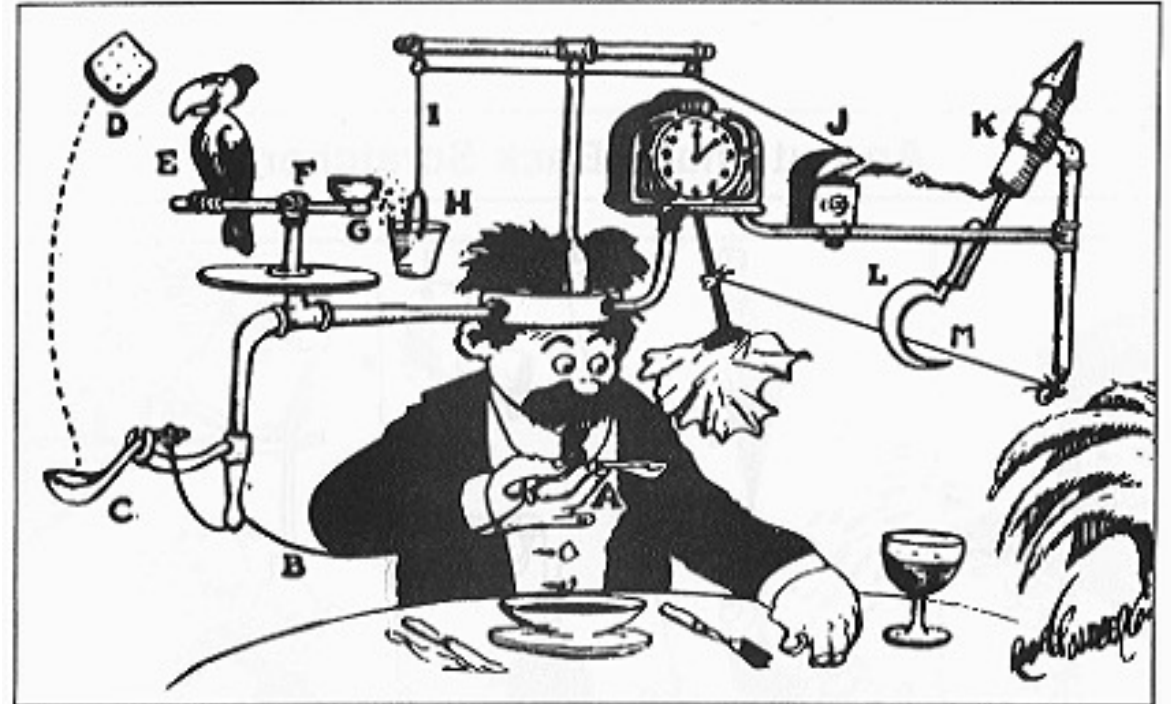
# Economy of Mechanism



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- Keep the design as simple as possible
  - Complex designs will be hard to analyze and verify

Self-Operating Napkin



# Fail-Safe (Fail-Close)



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- Default should be “no-access”
- Access should be based on explicit permissions
- When a system fails it should default to “no-access”



# Complete Mediation



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- Every access to every object should be checked for authorization
  - Enforces system-wide view of access control
  - Implies we need a fool-proof (secure) way to identify the source of every request
  - Caching access decisions should be avoided or done with care



# Open Design



The security of a design should not depend on the design being secret



Rather, design should be open



Security should depend on only specific information such as cryptographic keys, passwords being secret

# Separation-of-Privilege



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- Critical operations should be protected using two or more {keys, permissions, passwords} that can then be separated (e.g., 2-man rule)
  - One failure or compromise cannot bring down the system



# Least Privilege



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- Every program and every user of the system should operate using the least set of privileges necessary to complete the job
  - Limits the impact/damage of compromise

# Least Common Mechanism



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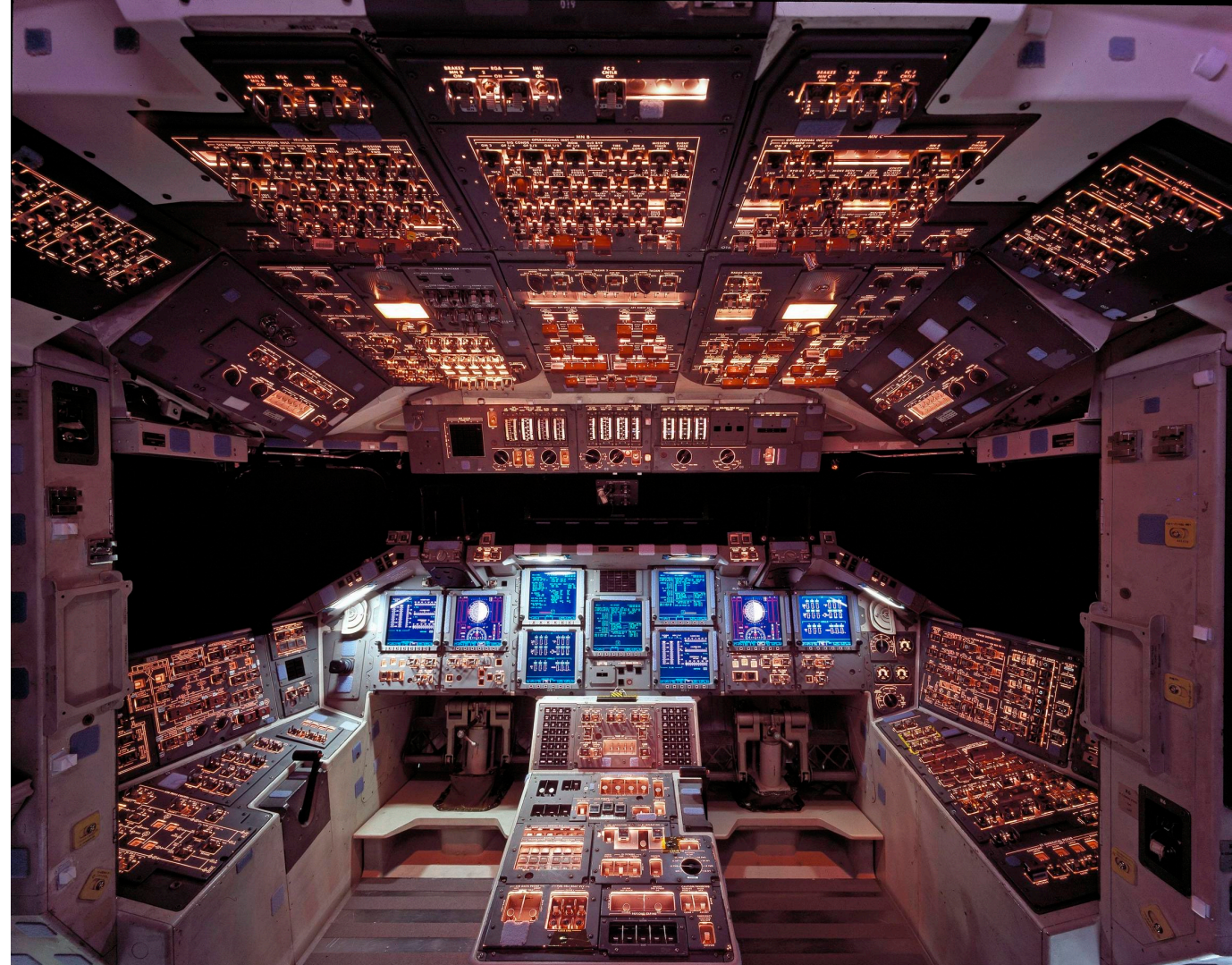
- Minimize the amount of mechanism common to more than one user and depended on by all users
  - shared mechanisms (e.g., shared libraries) may provide information paths from one user to another
  - could be a security risk

# Psychological Adoptability (Usability)



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- Essential that human interface be designed for ease of use
  - helps users routinely and automatically apply the protection mechanisms correctly



# Work Factor



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- Compare the cost of circumventing the mechanism with the resources of a potential attacker.



# Compromise Recording or Detection



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- mechanisms that reliably record that a compromise of information has occurred can be used in place of more elaborate mechanisms that completely prevent loss
  - argument for detection in place of protection/prevention

# Summary



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Economy of  
Mechanism (Keep  
it simple)

Fail-safe (fail-close)

Complete  
Mediation

Open Design

Separation-of-  
privilege

Least-privilege

Least common  
mechanism

Psychological  
acceptability  
(Usability)

Work Factor

Failure Recording or  
Detection



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# Cyber Security Strategy

# Security Strategy



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- Specification/Policy
  - What does it mean to be secured in particular?
- Implementation/Mechanism
  - How to enforce the specified security policy?
- Correctness/Assurance
  - Does the security system work as expected/advertised?
- Incentives
  - Are there right incentives for creating right policy, right mechanisms, and using them?



**“You must pinky-swear to never reveal our company secrets. That’s the cornerstone of our new information security program.”**

# Specification/Policy



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- Policy
  - A statement of what is and what is not allowed
  - Divides the world into secure and non-secure states
  - A secure system starts in a secure state. All transitions (changes in system state) keep it in a secure state
- Specification considerations
  - Security vs. ease of use
  - Return on investment – security business case

# Is this situation secure?



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- Web server accepts all connections
  - No authentication required
  - Self-registration
  - Connected to the Internet

# Security Mechanism



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A method, tool, or procedure for enforcing a security policy

- Prevention
  - e.g., encrypting data to preserve confidentiality
- Detection
  - e.g., using integrity checks like crypto hashes to detect tampering
- Response
  - e.g., locking out an account that had too many failed login attempts
- Recovery
  - e.g., backing up data to remote secure storage to recover from data loss

# Trust and Assumptions



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Locks prevent unwanted physical access

What are the assumptions this statement builds on?

# Assurance



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- Evidence of how much to trust a system
- Evidence can include
  - System specifications
  - Design
  - Implementation

# Aspirin Assurance Example



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- Why do you trust Aspirin from a major manufacturer?
  - FDA certifies the aspirin recipe
  - Factory follows manufacturing standards
  - Safety seals on bottles
- Analogy to software assurance

# Incentives



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- Incentives driving the policy
  - what should be allowed and what shouldn't be and why?
- Incentives driving mechanisms
  - should I use passwords or hardware-tokens?
- Incentives driving usage
  - what incentives are there for me to choose a strong password?  
for not writing it down?
- Incentives driving attackers
  - money? fame? ease of breaking in? fun?

# Summary



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- We examined the four prongs of a good security strategy
  - Security Policy
  - Security Mechanisms
  - Assurance or Correctness
  - Incentives



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# What is Cryptography

# What is Cryptography (1 of 2)



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Literally...

“Secret Writing”

- Historically focused on encryption
  - Ceaser Cipher is one well-known historical example

# What is Cryptography (2 of 2)



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- Today: much more than secret writing
- Tools & techniques for securing information
  - at rest
  - in transit
  - and even during computations
- A very important piece of security puzzle
  - albeit not a silver bullet

# Some Key Cryptographic Tools

## (1 of 2)



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- Encryption/Ciphers
  - support Confidentiality and Privacy
  - e.g., AES, 3DES, RSA, ElGamal
- Cryptographic Hashes
  - support Integrity
  - e.g., SHA-256, SHA-512

# Some Key Cryptographic Tools

## [2 of 2]



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- Message Authentication Codes (MACs)
  - support Integrity
  - e.g., HMAC-SHA256, AES-CBC-MAC
- Digital Signatures
  - support Integrity, Authenticity and Non-repudiation
  - e.g., RSA, DSS

# Summary



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- Cryptography was historically focused on encryption or 'secret writing'
- Today provides many essential tools beyond encryption for computer and data security



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# What is Encryption?

# What is Encryption? (1 of 3)



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Encryption primitives or Ciphers are cryptographic primitives used for preserving data confidentiality (and privacy)

# What is Encryption? (2 of 3)



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Encryption transforms 'plaintext' (comprehensible)  
data into 'ciphertext' (incomprehensible) data

HELLO WORLD ----> KHOOR ZRUOG

# What is Encryption? (3 of 3)



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Knowledge of the 'cryptographic key' used for the transformation is needed to reverse the transformation

# Types of Encryption



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Two main types for encryption

- Symmetric Encryption
  - Key used for enciphering (encryption) and deciphering (decryption) is the same
  - Hence the name symmetric (also called secret-key encryption)
  - Historical (classical) crypto is symmetric
- Asymmetric (or Public-Key) Encryption
  - Keys used for for enciphering (encryption) and deciphering (decryption) are related but different
  - Hence the name asymmetric (also called public-key encryption)
  - Asymmetric crypto is new

# Symmetric Encryption



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- Widely used; commonly called secret-key encryption
- Provides confidentiality
- Five components
  - Plaintext – original message
    - denoted by  $m$
  - encryption algorithm
    - denoted by  $E$
  - Shared secret key
    - denoted by  $K$
  - ciphertext – encrypted message
    - denoted by  $c$ ;  $c = E(m, k)$  or  $E_K(M)$  or  $c = \{m\}_K$
  - decryption algorithm
    - denoted by  $D$ ;
- Correctness
  - $m = D(E(m, K), K)$



Shared Secret  
Key



Plaintext  
(input)

Encryption



Ciphertext

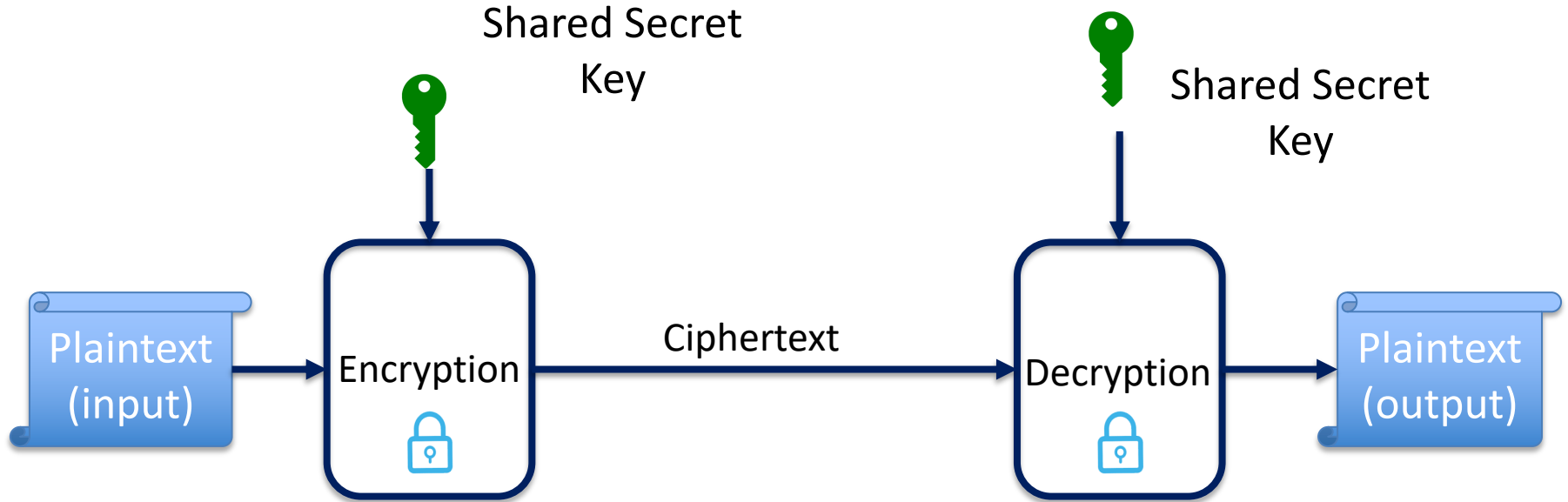


Shared Secret  
Key

Decryption



Plaintext  
(output)



# Public Key Cryptography



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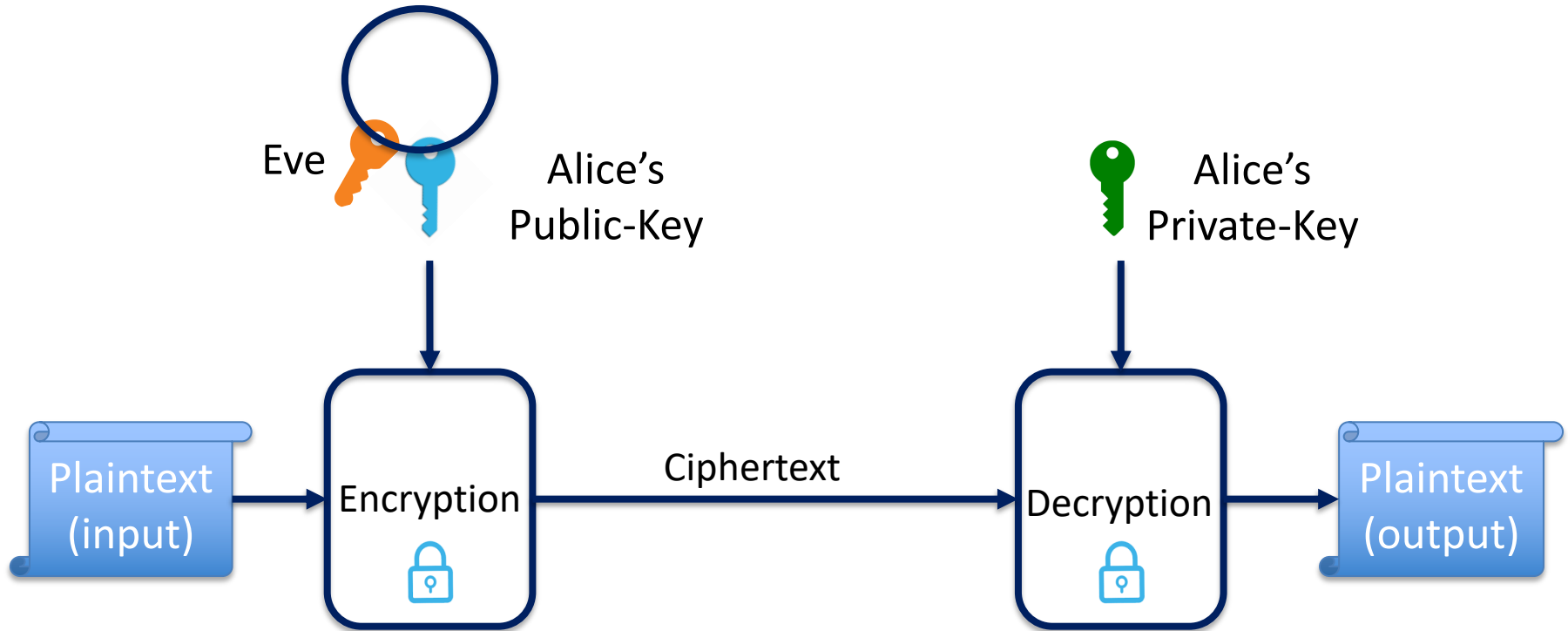
- Two keys
  - *Private key* known only to individual
  - *Public key* available to anyone
- Usage:
  - Confidentiality: encipher using public key, decipher using private key
  - Integrity/authentication: encipher using private key, decipher using public one
  - Symmetric Key distribution

# Asymmetric Encryption



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- Widely used; commonly called public-key encryption
- Provides confidentiality
- Five components
  - Plaintext – original message
    - denoted by  $m$
  - encryption algorithm
    - denoted by  $E$
  - key-pair: public-key, secret-key
    - denoted by  $(PK, SK)$
  - ciphertext – encrypted message
    - denoted by  $c$ ;  $c = E(m, PK)$  or  $E_{PK}(M)$  or  $c = \{m\}_{PK}$
  - decryption algorithm
    - denoted by  $D$ ;
- Correctness
  - $m = D(E(m, PK), SK)$



# Attacks on Ciphers [1 of 2]



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- Opponent whose goal is to break encryption (cryptosystem) is the *adversary*
- Standard cryptographic practice:
  - Assume adversary knows algorithm used, but not the key
  - **Also Known as : Kerckhoffs's principle or Shannon's Maxim**

# Attacks on Ciphers [2 of 2]



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## Types of attacks

- *ciphertext only*: adversary has only ciphertext; goal is to find plaintext, possibly key
- *known plaintext*: adversary has ciphertext, corresponding plaintext; goal is to find key
- *chosen plaintext*: adversary may supply plaintexts and obtain corresponding ciphertext (access to encryption oracle); goal is to find key
- *chosen ciphertext*: adversary may also supply ciphertexts and obtain corresponding plaintext (access to decryption oracle); goal is to find key

# Brute Force Attacks



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- Systematically checking all keys (pass-phrases) with the hope of eventually finding the right key
  - Also known as exhaustive search
- Example:
  - For *ciphertext only*: given a ciphertext try all possible keys to see if a comprehensible plaintext emerges
  - For *known plaintext*: given a ciphertext and plaintext pair try all possible keys to find the key that decrypts the ciphertext to the plaintext

# Basis for Attacks



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- Mathematical attacks
  - Based on analysis of underlying mathematics
- Statistical attacks
  - Make assumptions about the distribution of letters, pairs of letters (digrams), triplets of letters (trigrams), *etc.*
    - Called *models of the language*
  - Examine ciphertext, correlate properties with the assumptions

# Summary



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- Encryption transforms plaintext (comprehensible) into ciphertext (not comprehensible without the key)
- Two types of Encryption
  - Symmetric: same key for encryption and decryption
  - Asymmetric: different key for encryption and decryption
- Symmetric encryption has been around a long time
  - At least since the time of Caesar
- Asymmetric encryption is newer (publicly known since late 70's)
- Adversaries can try to break an encryption system
  - try to guess the plaintext by looking at the ciphertext
  - or guess the key after looking at one or more (plaintext, ciphertext) pairs



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# Classical Ciphers

# Classical Ciphers [1 of 3]



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- Encryption or ciphers have been around for a long time
  - At least since Caesar's time
  - Caesar Cipher is a well-known historical example



# Classical Ciphers [2 of 3]



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- Classical ciphers are ***symmetric***
  - Sender, receiver share common key
  - Keys may be the same, or trivial to derive from one another
- Two basic types
  - *Transposition* ciphers
  - *Substitution* ciphers
  - Combinations are called *product ciphers*

# Classical Ciphers [3 of 3]



- Transposition Ciphers
  - **Permute the symbols** (characters) in the message
  - Examples: Rail cipher, n-transpositional cipher, scytale
- Substitution Ciphers
  - **Substitute the symbols** (characters) in the message
  - Examples: Cæsar cipher, One Time Pad, Book cipher

# Transposition Cipher [1 of 3]



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- Plaintext and Ciphertext use the same alphabet
- Scramble letters in plaintext to produce ciphertext
- Example (Rail-Fence)
  - Plaintext is HELLO WORLD
  - Write the plaintext on alternating “rails”
    - »H E
    - »L L
    - »O W
    - »O R
    - »L D
  - Ciphertext is HLOOLELWRD

# Transposition Cipher [1 of 2]



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- Generalize to n-columnar transpositions
- Write text in fixed length rows.
- Read ciphertext out in column major order

H E L

L O W

O R L

D X X

HLODEORXLWLX

– Could also permute the columns

# Transposition Ciphers [1 of 3]



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- What is the key in Rail-Fence or n-columnar transposition?

# Other Transposition Ciphers



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- Scytale (older than 300BC in Greece!)





# Columnar Transposition

- ▶ # of columns and order of columns is encoded by a key
  - ▶ Ex: “WORD”  $\Rightarrow$  4 columns and rearranged in the order 4, 2, 3, 1
    - W is 4th alphabetically in the key
    - O is 2nd alphabetically in the key...

# Double Columnar Transposition



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- ▶ Apply Columnar Transposition twice
  - ▶ Can use two different passwords

# Double Columnar Transposition



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- ▶ Apply Columnar Transposition twice
  - ▶ Can use two different passwords
- ▶ Used in WWI US Military



# Double Columnar Transposition

- ▶ Apply Columnar Transposition twice
  - ▶ Can use two different passwords
- ▶ Used by US Military in WW-I
- ▶ Ubchi -- a variation used by Germany in WW-I
  - ▶ Same password twice
  - ▶ different padding

# Substitution Ciphers [1 of 2]



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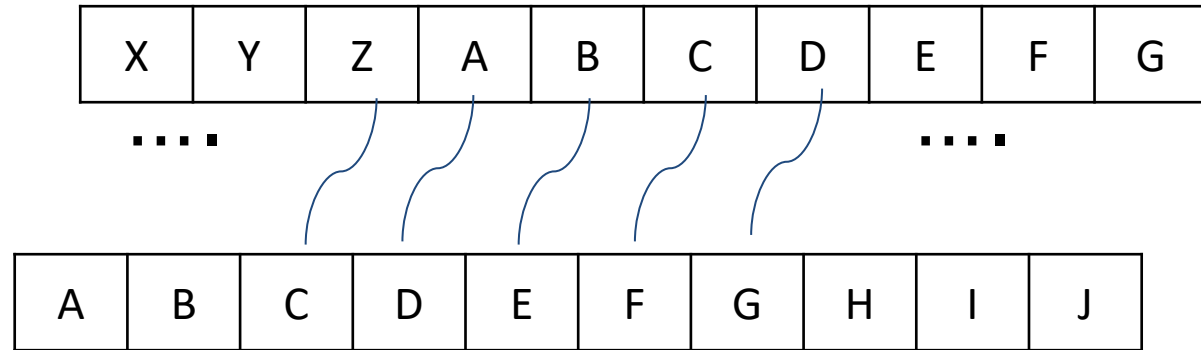
- Change characters in plaintext to produce ciphertext
- Example (Cæsar cipher)
  - Plaintext is HELLO WORLD
  - Change each letter to the third letter following it (A goes to D, X goes to A, Y to B, Z to C)
  - Ciphertext is KHOOR ZRUOG

# Caesar Cipher



Reportedly used by Julius Caesar

- It is a shift cipher with a shift of 3 (A → D)





# Caesar Cipher

Plaintext: Hello World

$H \rightarrow K, E \rightarrow H, L \rightarrow O, O \rightarrow R$   
 $\Rightarrow \text{Hello} \rightarrow \text{Khoor}$

$W \rightarrow Z, O \rightarrow R, R \rightarrow U, L \rightarrow O, D \rightarrow G$   
 $\Rightarrow \text{World} \rightarrow \text{Zruog}$

# Caesar Cipher



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It is a Substitution Cipher:

Plaintext Alphabet:

|   |   |   |   |   |   |   |   |   |   |     |
|---|---|---|---|---|---|---|---|---|---|-----|
| A | B | C | D | E | F | G | H | I | J | ... |
|---|---|---|---|---|---|---|---|---|---|-----|

Ciphertext Alphabet:

|   |   |   |   |   |   |   |   |   |   |     |
|---|---|---|---|---|---|---|---|---|---|-----|
| D | E | F | G | H | I | J | K | L | M | ... |
|---|---|---|---|---|---|---|---|---|---|-----|

# Keys in Ciphers



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- What is the “key” for Caesar Cipher?

# Caesar Cipher Exercise



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<http://rumkin.com/tools/cipher/>

Decode

Fdhvhu Vkliw Lv Grsh



# Other Substitution Ciphers

- ▶ Pigpen (Masonic or Tic-Tac-Toe) Cipher

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| A | B | C | J | K | L |
| D | E | F | M | N | O |
| G | H | I | P | Q | R |

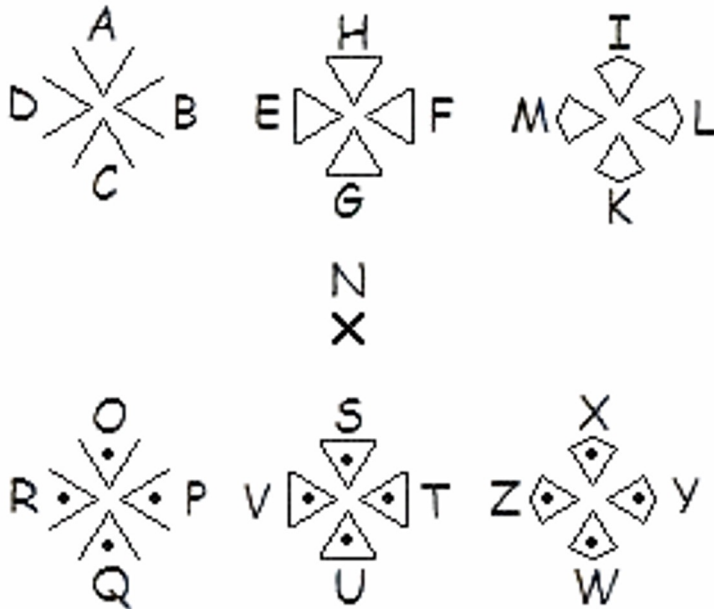
|   |   |   |
|---|---|---|
|   | S |   |
| T | X | Y |
|   | V | Z |





# Other Substitution Ciphers

## ► Templar Cipher



# Other Substitution Ciphers



## ► Rosicrucian Cipher

### ROSICRUCIAN CIPHER

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| A | B | C | D | E | F | G | H | I |
| . | . | . | . | . | . | . | . | . |
| J | K | L | M | N | O | P | Q | R |
| . | . | . | . | . | . | . | . | . |
| S | T | U | V | W | X | Y | Z |   |
| . | . | . | . | . | . | . | . |   |

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| K | = | ┐ | T | = | ┐ |
| N | = | ┐ | E | = | ┐ |
| I | = | ┐ | M | = | ┐ |
| G | = | ┐ | P | = | ┐ |
| H | = | ┐ | L | = | ┐ |
| T | = | ┐ | A | = | ┐ |
| S | = | ┐ | R | = | ┐ |



# Keys in Ciphers

- What is the “key” for Tic-Tac-Toe Cipher?



# Other Substitution Ciphers

- ▶ Caesar Cipher is “*monoalphabetic*”
  - ▶ *one* substitution alphabet
- ▶ What if we can combine two alphabets?
  - ▶ Ex: one substitution alphabet for odd letters and a different alphabet for one for even letters?
- ▶ What if we can use more than two!
  - ▶ we get “*polyalphabetic*” ciphers



# Polyalphabetic Ciphers

## Vigenere Cipher

- Key has multiple letters
  - Ex: "*passwd*"
    - Contrast with Caesar (Key is "*d*")



# Polyalphabetic Ciphers

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- Hello World with key "*passwd*"
  - Try it on Rumkin!



# Polyalphabetic Ciphers

## Vigenere Cipher

- Key has multiple letters
  - Ex: "*paswd*"
    - Contrast with Caesar (Key is "*d*")
- Hello World with key "*paswd*"
  - Try it on Rumkin!
    - Wedhr Lojhg



- Both Ceasar and n-columnar are easily breakable!
- Example: breaking Ceasar Cipher
  - Brute Force: try all possible (26) keys
  - Statistical: look at letter frequencies
    - “E” is the most common letter in English language
    - most common letter in ciphertext likely corresponds to “E”

# Cryptanalysis - Breaking Ciphers!



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*Abu Yusuf Al Kindi*, Arab mathematician, philosopher from 9th century AD is credited with discovering *frequency analysis*

# One-Time Pad



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- A cipher with a random key at least as long as the message
  - Provably unbreakable!!
  - Why?
- Let us consider a one-time pad scheme that uses Caesar Cipher

# One-Time Pad



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- Look at ciphertext `DXQR`.
- Equally likely to correspond to plaintext `DOIT` (key `AJIY`) and to plaintext `DONT` (key `AJDY`) and any other 4 letters
- Warning: keys *must* be random, or you can attack the cipher by trying to regenerate the key
  - Approximations, such as using pseudorandom number generators to generate keys, are *not* random



- Approximate one-time pad with book text
  - Sender and receiver agree on text to pull key from
  - Bible, Koran, Phone Book
- Problem is that book text is not random
  - Combine English with English
  - Can still perform language based statistical analysis

# Summary



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- Classic ciphers are symmetric
- They are either transposition (or permutation ciphers), substitution ciphers, or product ciphers (combination)
- These pen and paper classical ciphers have been used historically
- Not practical in the age of the computer – easy to break
- The components (transposition and substitution) are the same in modern ciphers
- One-time pad is provably unbreakable!